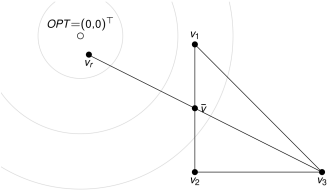
Optimization in Machine Learning

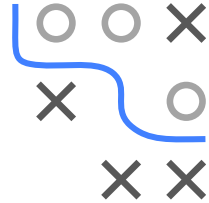
Nelder-Mead method



Learning goals

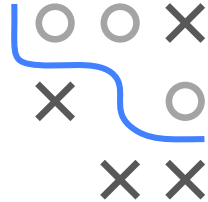
- General idea
- Reflection, expansion, contraction
- Advantages & disadvantages
- Examples

NELDER-MEAD METHOD



- Derivative-free method $\Rightarrow$ heuristic
- Generalization of bisection in d-dimensional space
- Based on d-simplex, defined by $d + 1$ points:
  - $d = 1$ interval
  - $d = 2$ triangle
  - $d = 3$ tetrahedron
  - $\cdots$

NELDER-MEAD



Advantages:

- No gradients needed
- Robust, often works well for non-differentiable functions.

Drawbacks:

- Relatively slow (not applicable in high dimensions)
- Not each step improves solution, only mean of corner values is reduced.
- No guarantee for convergence to local optimum / stationary point.

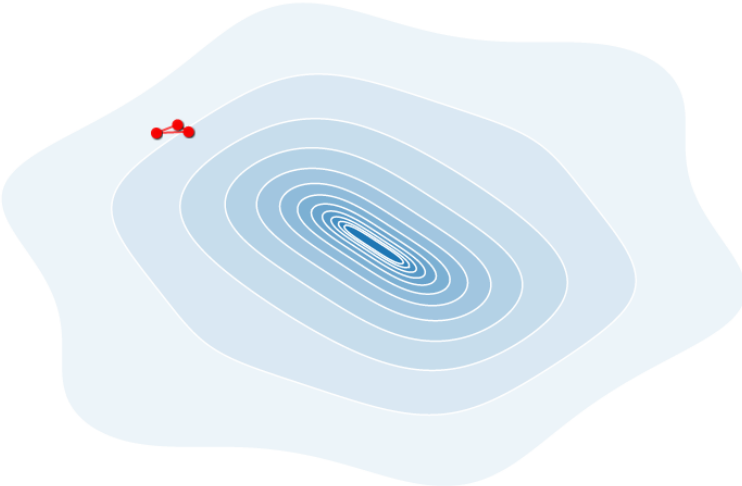
Visualization:

`http://www.benfrederickson.com/numerical-optimization/`

Note: Nelder-Mead is default method of `R` function `optim()`. If gradient is available and cheap, L-BFGS is preferred.

NELDER-MEAD VISUALIZATION IN 2D

$$\min_{\mathbf{x}} f(x_1, x_2) = x_1^2 + x_2^2 + x_1 \cdot \sin x_2 + x_2 \cdot \sin x_1$$



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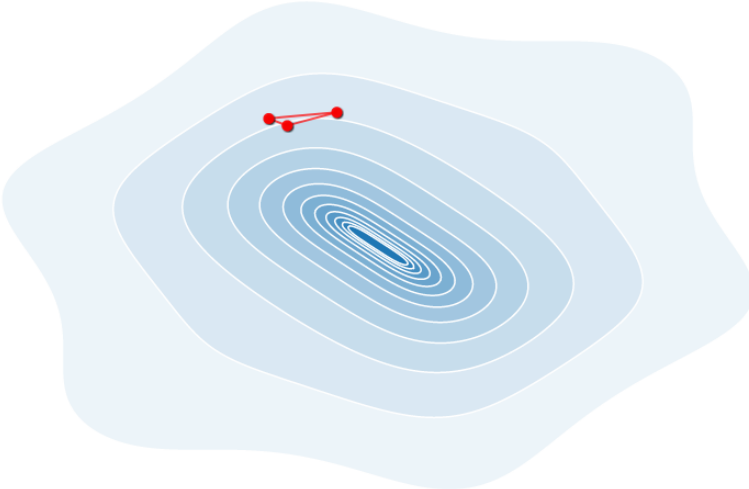

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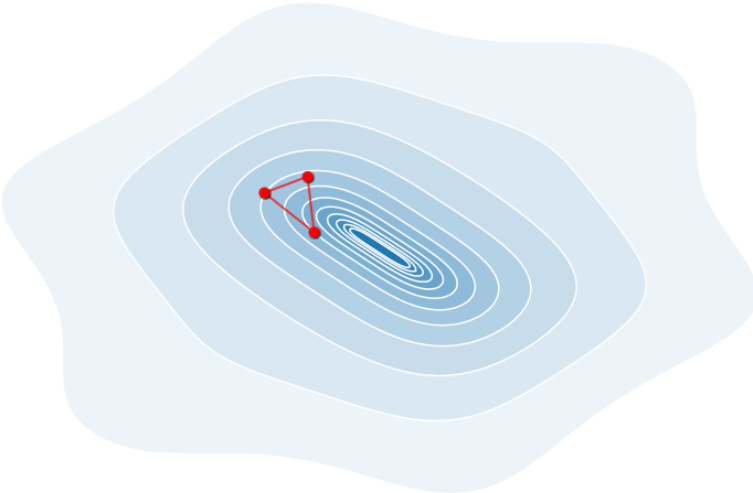

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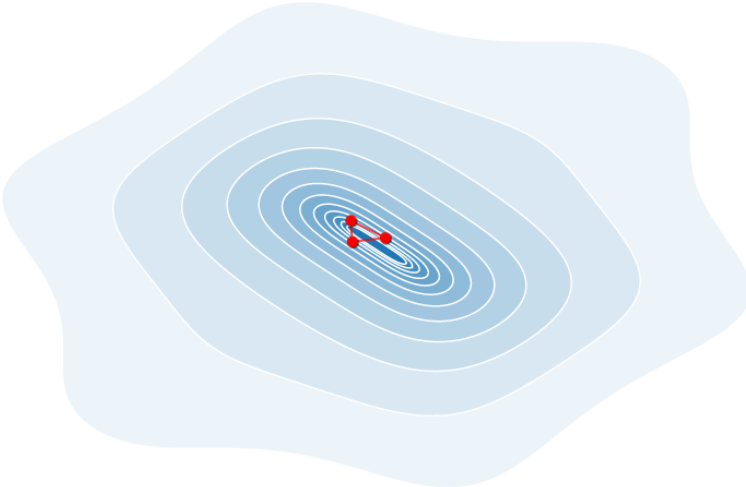
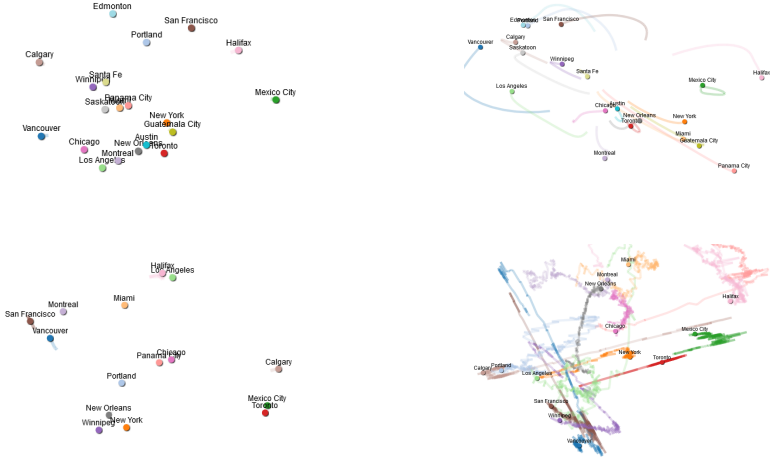


NELDER-MEAD VS. GD



Nelder-Mead in multiple dimensions: Organize points (US cities) to keep predefined mutual distances. For 10 cities, gradient descent (top) converges well for a suitable learning rate. Nelder-Mead (bottom) fails to converge, even after many iterations.